



# NORTH SYDNEY BOYS HIGH SCHOOL

2013 Year 11 Task 1

## Mathematics Extension 1

### General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

### Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Fletcher
- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Lucas
- ☐ Mrs Ziazaris

Student Name: \_\_\_\_\_

(To be used by the exam markers only.)

| Question No | 1         | 2         | 3         | 4         | 5         | Total     | Total %    |
|-------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| Mark        | <u>10</u> | <u>12</u> | <u>11</u> | <u>10</u> | <u>13</u> | <u>56</u> | <u>100</u> |

**Question 1 (10 marks)**     Start a new page. **Marks**

- (a) Evaluate  $\frac{7 \cdot 62 \times 10^3}{5 \cdot 169 \times 10^{-4}}$  giving your answer in scientific notation correct to two significant figures. 2
- (b) Write  $0.\dot{2}3\dot{4}$  as a fraction in lowest terms, showing all working. 2
- (c) Simplify  $3\sqrt{32} + 2\sqrt{50} - 8\sqrt{18}$  2
- (d) If  $(a + \sqrt{3})^2 = b + 10\sqrt{3}$ , where  $a$  and  $b$  are integers, then find the value of  $a$  and  $b$ . 2
- (e) If  $x = \sqrt{2} - 1$  then show that  $x - \frac{1}{x}$  is rational. 2

**Question 2 (12 marks)**     Start a new page.

- (a) Factorise the following expressions:
- (i)  $6x^2 - 5x - 21$  2
- (ii)  $500 + 4x^3$  2
- (iii)  $1 - a^7$  2
- (iv)  $x^2 - 9y^2 + 7x + 21y$  2
- (b) Simplify the following:
- (i)  $\frac{7-x}{3} - \frac{3-4x}{8}$  2
- (ii)  $\frac{6^n}{2^{n+1} + 2^n}$  2

**Question 3 (11 marks)**     Start a new page.  
Solve the following equations and inequalities:

- (a)  $(3x + 2)^2 = 25$  2
- (b)  $3^{-2x} = \frac{1}{27}$  2
- (c)  $|3x + 4| = 10$  2
- (d)  $|2x - 5| > 7$  2
- (e)  $|x - 3| = 4x + 2$  3

**Question 4 (10 marks)** Start a new page.

Sketch the graphs of the following relations. Use separate number planes and show all important features:

- (a)  $y = -\sqrt{9 - x^2}$  2
- (b)  $y = |2x - 1|$  2
- (c)  $y = 1 - 2^x$  2
- (d)  $y = \frac{x^3 - x}{x^2 - 1}$  2
- (e)  $y^2 = x - 9$  2

**Question 5 (13 marks)** Start a new page.

- (a) Prove that the function  $f(x) = x^4 - 2x^2 + 3$  is even. 2
- (b) A function  $f(x)$  is defined piece-wise by the rule:
- $$f(x) = \begin{cases} 4, & x \leq 0 \\ (x-2)^2, & 0 < x \leq 3 \\ 1, & x > 3 \end{cases}$$
- (i) Evaluate  $f(2) + f[f(-2)]$  2
- (ii) What is the range of  $f(x)$ ? 1
- (c) Find  $f(x)$  if  $f(x-1) = 2x + 9$ . 1
- (d) Find the domain and range of  $y = \sqrt{x+1} + \sqrt{x-1}$ . 2
- (e) (i) Sketch the graph of  $y = \frac{x-1}{x+1}$  showing any asymptotes and intercepts. 3
- (ii) Using your sketch, or otherwise, solve the inequality  $|x+1| > \frac{x-1}{x+1}$ . 2

END OF EXAMINATION

Q1.

$$(a) \frac{7.62 \times 10^3}{5.169 \times 10^{-4}} = 1.474\,172\,95 \times 10^7$$

$$= 1.5 \times 10^7 \quad (2 \text{ sig. figs.})$$

✓✓

$$(b) \text{ Let } x = 0.2\dot{3}\dot{4}$$

$$\therefore x = 0.2\,34\,34\,34\ldots$$

$$100x = 23.4\,34\,34\,34\ldots$$

✓ method

$$\text{Thus } 99x = 23.2$$

$$x = \frac{232}{990}$$

$$= \frac{116}{495}$$

✓ correct answer

$$(c) \quad 3\sqrt{32} + 2\sqrt{50} - 8\sqrt{18} = 12\sqrt{2} + 10\sqrt{2} - 24\sqrt{2}$$

$$= -2\sqrt{2}$$

✓

✓

$$(d) \quad (a + \sqrt{3})^2 = b + 10\sqrt{3}$$

$$\therefore a^2 + 3 + 2a\sqrt{3} = b + 10\sqrt{3}$$

$$\therefore 2a = 10 \quad \therefore a = 5$$

$$\therefore b^2 = 25 + 3 = 28$$

$$\text{Thus } a = 5$$

$$b = 28$$

✓✓

$$(e) \quad x = \sqrt{2} - 1$$

$$\therefore x - \frac{1}{x} = \sqrt{2} - 1 - \frac{1}{\sqrt{2} - 1}$$

$$= \sqrt{2} - 1 - \frac{1}{\sqrt{2} - 1} \cdot \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$= \sqrt{2} - 1 - (\sqrt{2} + 1)$$

$$= -2, \text{ which is rational.}$$

✓

✓

**Q2**

(a)

$$(i) \quad 6x^2 - 5x - 21 = (3x - 7)(2x + 3)$$

✓  
✓ (one for each factor)

$$(ii) \quad 500 + 4x^3 = 4(125 + x^3) \\ = 4(5 + x)(25 - 5x + x^2)$$

✓✓

$$(iii) \quad 1 - a^7 = (1 - a)(1 + a + a^2 + a^3 + a^4 + a^5 + a^6)$$

✓✓

$$(iv) \quad x^2 - 9y^2 + 7x + 21y = (x - 3y)(x + 3y) + 7(x + 3y) \\ = (x + 3y)(x - 3y + 7)$$

✓

✓

(b)

$$(i) \quad \frac{7-x}{3} - \frac{3-4x}{8} = \frac{8(7-x) - 3(3-4x)}{24} \\ = \frac{56 - 8x - 9 + 12x}{24} \\ = \frac{4x + 47}{24}$$

✓

✓

$$(ii) \quad \frac{6^n}{2^{n+1} + 2^n} = \frac{2^n \cdot 3^n}{2^n(2^1 + 1)} \\ = \frac{3^n}{3} \\ = 3^{n-1}$$

✓ for correct numerator or denominator

✓

**Q3**

(a)  $(3x+2)^2 = 25$

$\therefore 3x+2 = 5 \text{ or } -5$

$\therefore 3x = 3 \text{ or } -7$

$\therefore x = 1 \text{ or } -\frac{7}{3}$

✓✓

(b)  $3^{-2x} = \frac{1}{27} = 3^{-3}$

$\therefore -2x = -3 \quad \therefore x = \frac{3}{2}$

✓✓

(c)  $|3x+4| = 10$

$\therefore 3x+4 = 10 \text{ or } 3x+4 = -10$   
 $3x = 6 \quad 3x = -14$   
 $x = 2 \quad x = -\frac{14}{3}$

$\therefore x = 2 \text{ or } -\frac{14}{3}$

✓✓

(d)  $|2x-5| > 7$

$\therefore 2x-5 > 7$   
 $2x > 12$   
 $x > 6$

or  $2x-5 < -7$   
 $2x < -2$   
 $x < -1$

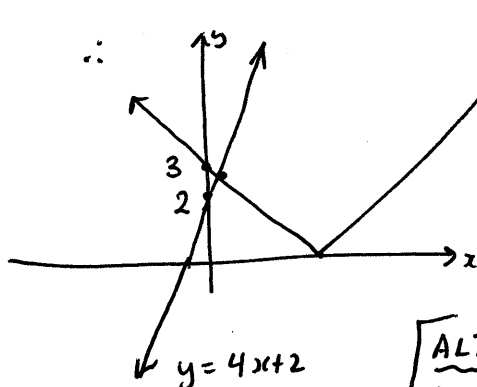
Thus:

$x > 6 \text{ or } x < -1$

✓

✓ (penalise "and")

(e)  $|x-3| = 4x+2$



$y = |x-3|$   
 Clearly, only 1 solution.  
 $\therefore$  We have  
 $4x+2 = -x+3$   
 $5x = 1$

$\therefore x = \frac{1}{5}$

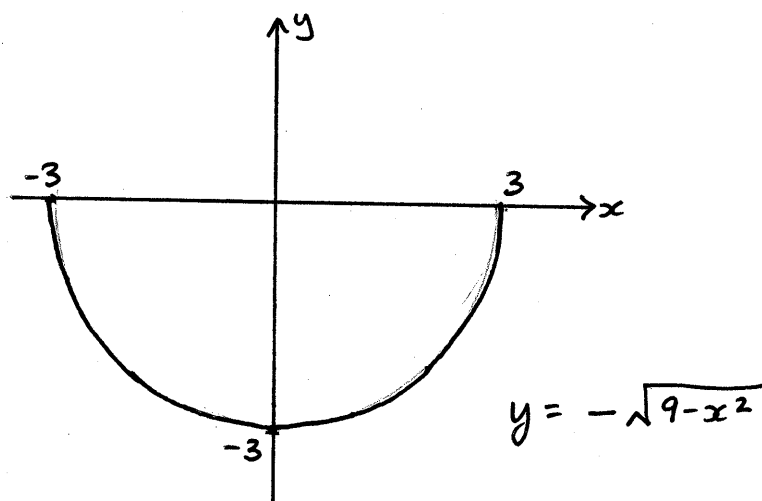
✓✓✓

[ALT: Candidates may go via  
 $x-3 = 4x+2$  or  $x-3 = -4x-2$ ,  
 but must test & discard  $x = -5/3$ ]

(needs correct working for full marks).

Q4

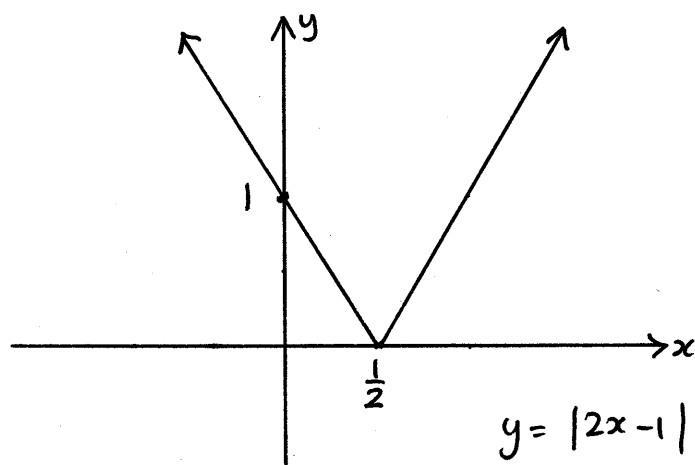
(a)



(\*) delete 1 mark (total) if axes unlabelled.

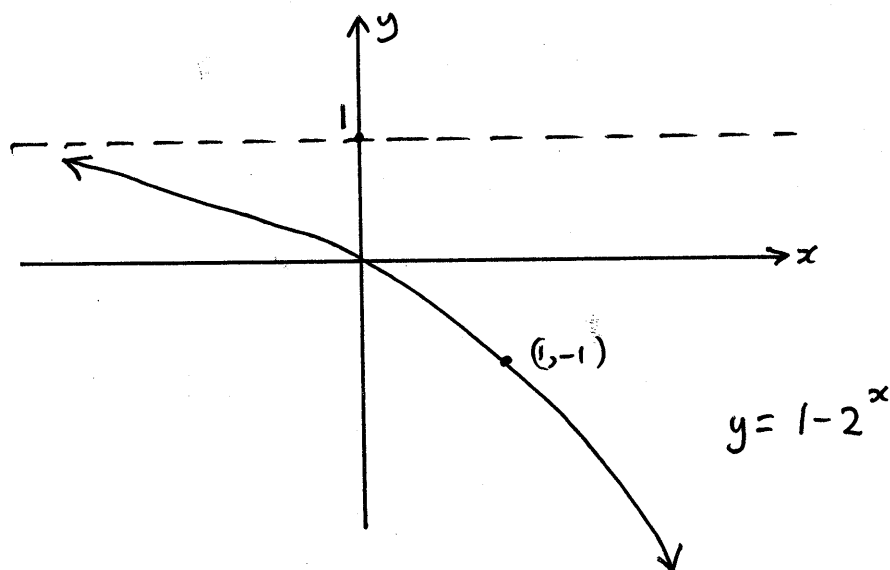
✓ shape/curve  
✓ intercepts

(b)



✓ shape  
✓ intercepts

(c)

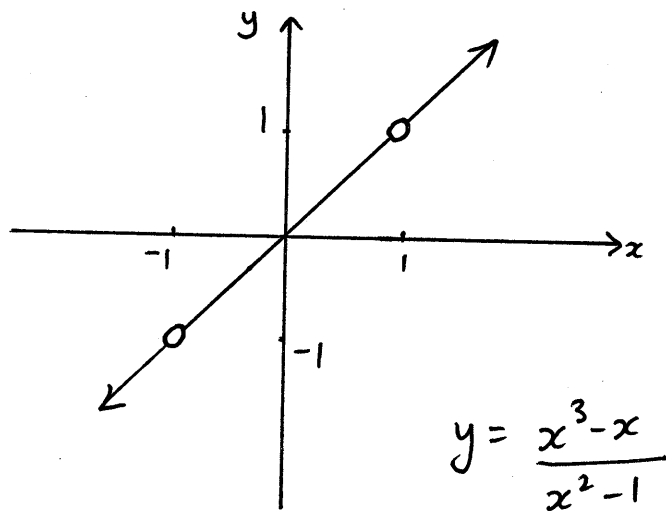


✓ curve  
✓ asymptote, point

Q4-continued

$$(d) \ y = \frac{x^3 - x}{x^2 - 1} = \frac{x(x^2 - 1)}{x^2 - 1}$$

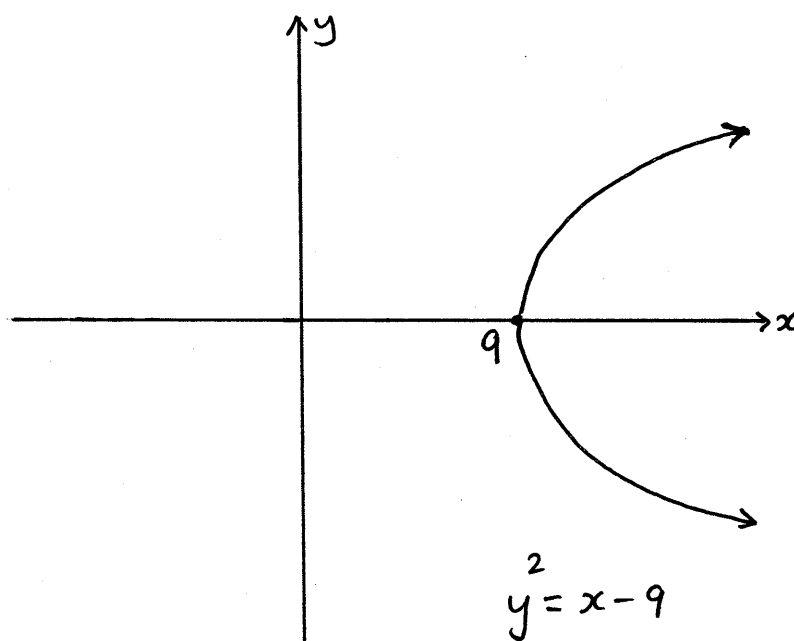
$$\therefore y = x \quad (x \neq -1, 1)$$



✓ 'curve'

✓ holes

(e)



✓ curve

✓ intercept

Q5

$$(a) \quad f(x) = x^4 - 2x^2 + 3$$

$$\begin{aligned} \therefore f(-x) &= (-x)^4 - 2(-x)^2 + 3 \\ &= x^4 - 2x^2 + 3 \\ &= f(x) \end{aligned}$$

$\therefore f(x)$  is even.

✓✓ correct demonstration

$$\begin{aligned} (b) \quad (i) \quad f(2) + f[f(-2)] &= (2-2)^2 + f(4) \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

✓

✓

$$(ii) \quad R: 0 \leq y \leq 4$$

✓

$$\begin{aligned} (c) \quad f(x-1) &= 2x + 9 \\ &= 2(x-1) + 11 \end{aligned}$$

$$\therefore f(x) = 2x + 11$$

✓

$$(d) \quad y = \sqrt{x+1} + \sqrt{x-1}$$

$$D: x \geq 1$$

✓

$$R: y \geq \sqrt{2}$$

✓

[P.T.O. →]

Q5 - continued

$$(e) (i) \quad y = \frac{x-1}{x+1} = \frac{x+1-2}{x+1}$$

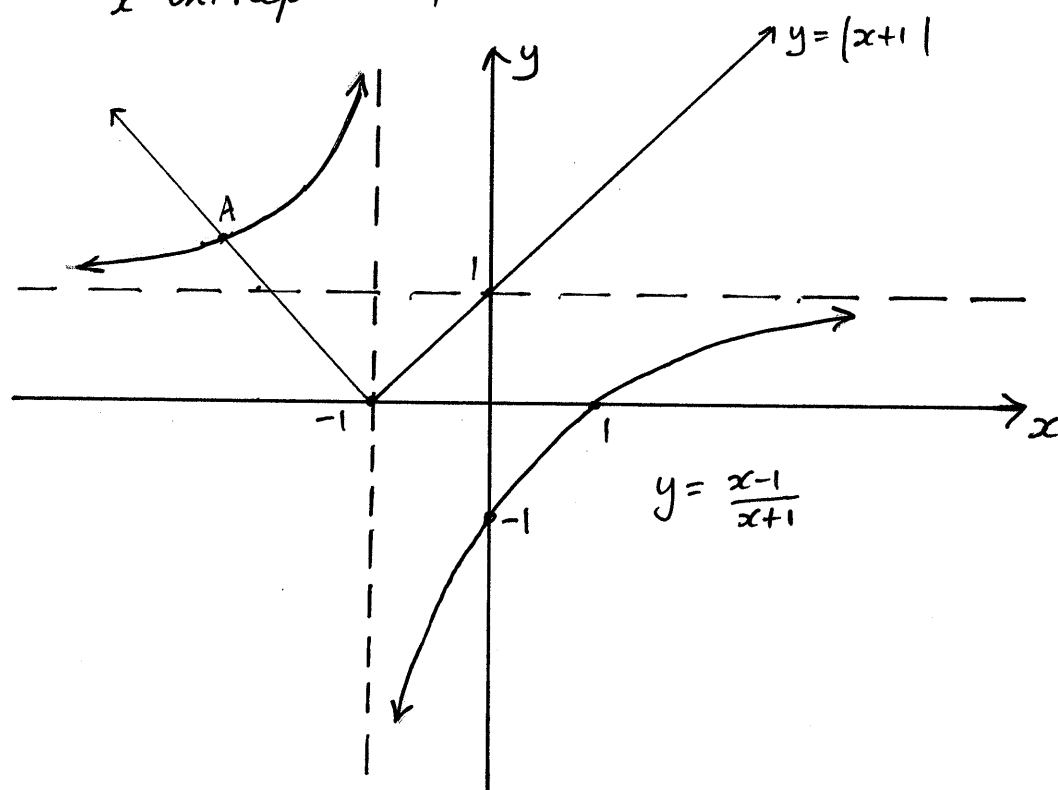
$$\therefore y = 1 - \frac{2}{x+1}$$

Thus, vertical asymptote is  $x = -1$

horizontal asymptote is  $y = 1$

y-intercept is  $-1$

x-intercept is  $1$



✓ curve  
✓ asymptotes  
✓ intercepts

(ii) By graphing  $y = |x+1|$  on the same number plane, we see we need to find point A:  $-x-1 = \frac{x-1}{x+1}$

$$\therefore -x^2 - 2x - 1 = x - 1$$

$$x^2 + 3x = 0 \quad \therefore x = -3 \quad [\text{from the diagram}]$$

$$\therefore \text{Solution to } |x+1| > \frac{x-1}{x+1}$$

is therefore  $x < -3$  or  $x > -1$

✓✓